Training Management System - TypeScript Assignment
Bug Analysis Report

Bug Number	Problem	Root Cause	Resolution
Bug-01	RecentlyViewedCourses.pop() throws a TypeScript error or causes runtime failures if the stack is empty.	Array.pop() can return undefined, but the method signature strictly specifies string as the return type.	Changed the method return type from string to string | undefined. Added a null check before calling .toUpperCase().
Bug-02	DoubtQueue.dequeue() throws a TypeScript error or causes runtime failures if the queue is empty.	Array.shift() can return undefined, but the method signature strictly specifies string.	Changed the method return type from string to string | undefined. Modified the console log to match the expected output instead of printing the string length.
Bug-03	traineeId + 10 outputs "10210" instead of 112.	traineeId was assigned as a string ("102"). Adding a number to a string causes string concatenation rather than mathematical addition.	Changed traineeId initialization from the string "102" to the number 102.